11.2 Consider a supply and demand model written in its most general implicit form, using capital Greek letters for the unknown parameters and E_i for the random errors,

$$\text{Demand: } \Gamma_{11}q + \Gamma_{21}p + \mathrm{B}_{11} + \mathrm{B}_{21}x + E_1 = 0$$
$$\text{Supply: } \Gamma_{12}q + \Gamma_{22}p + \mathrm{B}_{12} + \mathrm{B}_{22}x + E_2 = 0$$

a. Multiply each equation by 3. Do they remain true?

b. Multiply the demand equation by $-1/\Gamma_{11}$. Does it remain true?

c. Define $\alpha_{21} = -\Gamma_{21}/\Gamma_{11}$, $\beta_{11} = -\mathrm{B}_{11}/\Gamma_{11}$, $\beta_{21} = -\mathrm{B}_{21}/\Gamma_{11}$, $e_1 = -E_1/\Gamma_{11}$ and write the demand equation with q on the left-hand side and the remaining terms on the right-hand side. By choosing q to be on the left-hand side of the equation, we have chosen a **normalization rule**.

d. Repeat the process for the supply equation, beginning by multiplying through by $-1/\Gamma_{22}$, and obtain the normalized supply curve with

$$\alpha_{12} = -\Gamma_{12}/\Gamma_{22}, \ \beta_{12} = -\mathrm{B}_{12}/\Gamma_{22}, \ \beta_{22} = -\mathrm{B}_{22}/\Gamma_{22}, \ \text{and} \ e_2 = -E_2/\Gamma_{22}$$

Write the normalized supply equation with p on the left-hand side and the remaining terms on the right side.

e. Mathematically, in a system of jointly determined variables, it does not matter which variable appears on the left side of each normalized equation. True or false?

11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for ln(*PRICE*), adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?

b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.

c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

(a)

$$\ln(QUAN_t) = \beta_1 + \beta_2 \ln(PRICE_t) + \beta_3 STORMY_t + e_{st} \qquad (11.14)$$

$$\ln(PRICE_t) = \pi_{12} + \pi_{22}MON_t + \pi_{32}TUE_t + \pi_{42}WED_t + \pi_{52}THU_t + \pi_{62}STORMY_t + v_{t2} \qquad (11.16)$$

```
Call:
lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed,
    data = fultonfish)

Residuals:
     Min       1Q   Median       3Q      Max
-0.92281 -0.18819  0.00813  0.22404  0.68381

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.35957    0.07915  -4.543 1.50e-05 ***
mon         -0.10825    0.10339  -1.047  0.29753
tue         -0.06609    0.10103  -0.654  0.51446
wed         -0.04927    0.10377  -0.475  0.63591
thu          0.03935    0.10071   0.391  0.69681
stormy       0.44634    0.07921   5.635 1.51e-07 ***
mixed        0.23688    0.07851   3.017  0.00321 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3413 on 104 degrees of freedom
Multiple R-squared:  0.245,     Adjusted R-squared:  0.2014
F-statistic: 5.624 on 6 and 104 DF,  p-value: 4.349e-05
```

0.00321 < 0.05，故 mixed 在 5% 的顯著水準下統計顯著

```
> f_test <- linearHypothesis(model_rf, c("stormy = 0", "mixed = 0"))
> print(f_test)

Linear hypothesis test:
stormy = 0
mixed = 0

Model 1: restricted model
Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

  Res.Df    RSS Df Sum of Sq     F    Pr(>F)
1    106 15.876
2    104 12.115  2    3.7604 16.14 7.857e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

虛無假設：stormy = 0 且 mixed = 0（即天氣變數對價格沒有影響）

檢定結果： 7.857 *10^(-7)

結論：在 5% 的顯著水準下，我們拒絕虛無假設，即 stormy 和 mixed 具聯合顯著性

F 統計量為 16.14，大於 10

(b)

```
Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
    tue + wed + thu + stormy + mixed, data = fultonfish)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1327 -0.4320  0.0654  0.5133  1.1798

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533  < 2e-16 ***
lprice      -0.93014    0.35340  -2.632  0.00977 **
mon         -0.01190    0.20837  -0.057  0.95456
tue         -0.52583    0.20230  -2.599  0.01068 *
wed         -0.56262    0.20696  -2.719  0.00767 **
thu          0.09987    0.20285   0.492  0.62350

Diagnostic tests:
                 df1 df2 statistic  p-value
Weak instruments   2 104    16.140 7.86e-07 ***
Wu-Hausman         1 104     1.489    0.225
Sargan             1  NA     0.772    0.380
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared: 0.185,	Adjusted R-squared: 0.1462
Wald test: 4.927 on 5 and 105 DF,  p-value: 0.0004317
```

TABLE 11.5 2SLS Estimates for Fish Demand

Variable	Coefficient	Std. Error	*t*-Statistic	Prob.
C	8.5059	0.1662	51.1890	0.0000
ln(*PRICE*)	−1.1194	0.4286	−2.6115	0.0103
MON	−0.0254	0.2148	−0.1183	0.9061
TUE	−0.5308	0.2080	−2.5518	0.0122
WED	−0.5664	0.2128	−2.6620	0.0090
THU	0.1093	0.2088	0.5233	0.6018

本次 lprice 的標準誤為 0.35340，而 Table 11.5 的標準誤為 0.4286，顯示加入 MIXED 作為第二個工具變數後，價格係數的標準誤降低。

(c)

Sargan 檢定結果：p-value = 0.380

因為 p-value > 0.05，我們無法拒絕工具變數是有效的虛無假設，代表 STORMY 和 MIXED 作為工具變數是合理且有效，模型不存在過度識別（Over-identification）的問題。

(d)

```
> linearHypothesis(model_rf, c("mon = 0", "tue = 0", "wed = 0", "thu = 0"))

Linear hypothesis test:
mon = 0
tue = 0
wed = 0
thu = 0

Model 1: restricted model
Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    108 12.408
2    104 12.115  4   0.29256 0.6279 0.6437
```

mon, tue, wed, thu 的聯合 F 檢定結果為 0.6279，p-value 為 0.6437

顯示在 5% 的顯著水準下，我們無法拒絕虛無假設 (所有星期變數係數均為 0)，表示 MON, TUE, WED, THU 這些代表這些驅動需求的變數對價格沒有顯著的解釋能力。

要識別（Identify）供給方程，我們必須找到能夠引起需求曲線移動（但本身不影響供給）且能顯著影響價格的工具變數。然而檢定結果顯示，「星期幾」這些需求因子與價格之間並沒有顯著的相關性，表示這些變數是弱工具變數。在這種情況下，使用 2SLS 估計出的供給方程會不精確，甚至產生偏誤。

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

c. Estimate the price elasticity of demand "at the means" using the results from (b).

d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI, PS, and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.

e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

(a)

Demand: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$

$\Rightarrow \alpha_2 P_i = Q_i - (\alpha_1 + \alpha_3 PS_i + \alpha_4 DI_i + e_{di})$

$$P_i = \frac{-\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q_i - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di}$$

$$= r_1 + r_2 Q_i + r_3 PS_i + r_4 DI_i + u$$

if $r_2 < 0$, law of demand

$r_3 > 0$, substitate goods

$r_4 > 0$, truffles are normal goods

Supply: $\beta_2 P_i\ Q_i - (\beta_1 + \beta_3 PF_i + e_{si})$

$$\Rightarrow P_i = \frac{-\beta_1}{\beta_2} + \frac{1}{\beta_2} Q_i - \frac{\beta_3}{\beta_2} PF_i + \frac{1}{\beta_2} e_{si}$$

$$= \delta_1 + \delta_2 Q_i + \delta_3 PF_i + \varepsilon$$

if $\delta_2 > 0$, supply curve slope > 0

$\delta_3 > 0$, cost $\uparrow \Rightarrow P \uparrow$

(b) demand.

```
Call:
ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)

Residuals:
    Min      1Q  Median      3Q     Max 
-39.661  -6.781   2.410   8.320  20.251 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -11.428     13.592  -0.841  0.40810    
q             -2.671      1.175  -2.273  0.03154 *  
ps             3.461      1.116   3.103  0.00458 ** 
di            13.390      2.747   4.875 4.68e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567,	Adjusted R-squared: 0.5056 
Wald test: 17.37 on 3 and 26 DF,  p-value: 2.137e-06 
```

supply.

```
Call:
ivreg(formula = p ~ q + pf | pf + ps + di, data = truffles)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.7983 -2.3440 -0.6281  2.4350 11.1600 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -58.7982     5.8592  -10.04 1.32e-10 ***
q             2.9367     0.2158   13.61 1.32e-13 ***
pf            2.9585     0.1560   18.97  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486,	Adjusted R-squared: 0.9448 
Wald test: 232.7 on 2 and 27 DF,  p-value: < 2.2e-16 
```

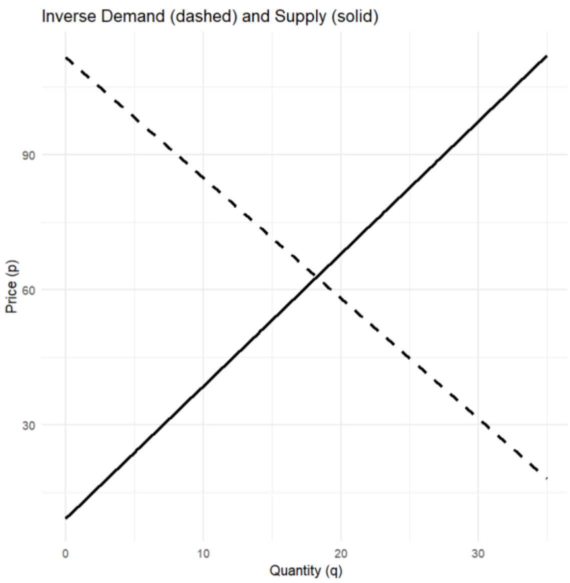
Using iverg( ) (2SLS) to estimate the demand and supply equations for truffles, the signs of all key variables are consistent with economic theory, including a downward-sloping demand curve, an upward-sloping supply curve, and the expected effects of substitute price, income, and production cost on price.

Moreover, except for the intercept term in the demand equation, all main explanatory variables are statistically significant at the 5% level (and even of the 1% or 0.1% level), indicating strong statistical significance in the model estimates.

(c)

```
> with(truffles, {
+     Pbar <- mean(p);    Qbar <- mean(q)
+     alpha2 <- 1 / coef(demand.iv)["q"]
+     elas  <- alpha2 * Pbar / Qbar
+     round(elas, 3)
+ })
     q
-1.272
```

(d)



(e)

```
> c(p_star = round(p_star, 2), q_star = round(q_star, 2))
p_star.(Intercept) q_star.(Intercept)
             62.84              18.25
```

The reduced-form model predicts a quantity of 18.26 and a price of 62.84. Based on this comparison, we believe the results from both approaches are highly consistent.

(f)

demand.

```
Call:
lm(formula = p ~ q + ps + di, data = truffles)

Residuals:
     Min       1Q   Median       3Q      Max 
-25.0753  -2.7742  -0.4097   4.7079  17.4979 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -13.6195     9.0872  -1.499   0.1460    
q             0.1512     0.4988   0.303   0.7642    
ps            1.3607     0.5940   2.291   0.0303 *  
di           12.3582     1.8254   6.770 3.48e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared:  0.8013,	Adjusted R-squared:  0.7784 
F-statistic: 34.95 on 3 and 26 DF,  p-value: 2.842e-09
```

supply.

```
Call:
lm(formula = p ~ q + pf, data = truffles)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.4721 -3.3287  0.1861  2.0785 10.7513 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -52.8763     5.0238  -10.53 4.68e-11 ***
q             2.6613     0.1712   15.54 5.42e-15 ***
pf            2.9217     0.1482   19.71  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared:  0.9531,	Adjusted R-squared:  0.9496 
F-statistic: 274.4 on 2 and 27 DF,  p-value: < 2.2e-16
```

With the exception of the demand equation estimated using OLS-where the sign of the quantity coefficient is incorrect -all other coefficient signs are as expected. Furthermore, aside from the intercept and the coefficient on quantity in the OLS demand model, all other coefficients are statistically significant at conventional levels.